

Advanced Partial Fraction Decomposition Practice

Instructions: Decompose each rational function into partial fractions. Simplify where necessary.

1.

$$\frac{x^4 + 3x^2 + 2}{(x^2 + 1)(x + 1)}$$

2.

$$\frac{3x^2 + 5x + 4}{(x - 2)^2(x^2 + 1)}$$

3.

$$\frac{2x^2 + 7x + 6}{x^3 + x^2 - x - 1}$$

4.

$$\frac{3x^4 + x^3 + 5x^2 + 4x + 7}{(x^2 + 1)^2}$$

5.

$$\frac{x^5 + 2x^3 + x + 1}{(x^2 + 1)^2(x - 1)}$$

6.

$$\frac{5x^2 + 3x + 2}{x^3 + 2x^2 + x}$$

7.

$$\frac{2x^2 + 3x + 1}{x^3 - x}$$